

AWAIS ASGHAR

AI Engineer

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RESEARCH INTERESTS

Machine Learning, Deep Learning, Computer Vision, Generative AI, LLMs, RAG, Data Science, Agentic AI.

EDUCATION

National University of Sciences and Technology (NUST)

Islamabad, Pakistan

Bachelor of Engineering in Electrical Engineering

Sept 2022 - June 2026

CGPA: 3.63/4.0 | Specialization GPA: 3.94/4.0 | **Merit Scholarship: 2024-2026**

Senior Design Thesis: ML-based Hardware Accelerator for Real-Time Image Segmentation on FPGA

PAPERS IN PROGRESS

[In Progress] Design of a Novel Lightweight U-Net Architecture for Efficient Semantic Segmentation

- Based on FYP and autonomous driving segmentation research at NUST SEecs. Proposes a compressed encoder-decoder model reduced from 8.7M to 0.53M parameters, achieving 72% IoU and 96% test accuracy on the CARLA simulator dataset.

[In Progress] Real-Time Fabric Defect Detection on Edge Devices Using Classical Computer Vision

- Multi-method detection pipeline (GLCM, FFT, Gabor filters, statistical variance, IoU-based bounding box fusion) deployed on Jetson Nano for interpretable, low-power defect localization without deep learning.

RESEARCH EXPERIENCE

Mitacs Globalink Research Intern

Victoria, Canada | Summer 2026

NRC Herzberg Institute of Astrophysics | Supervisor: Sebastien Fabbro

- Conducting applied ML and AI research on large-scale astrophysical datasets, contributing to model development, scientific computing workflows, and data analysis pipelines.
- Gaining end-to-end academic research experience including literature review, experiment design, result analysis, and scientific documentation.

Deep Learning Research Intern

SINES, NUST | June 2024 - Sept 2024

Supervisor: Dr. Jameel Nawaz Malik, Assistant Professor, Deep Learning Lab SINES

- Built and evaluated deep neural network models (CNNs, transfer learning) using PyTorch and TensorFlow with GPU-accelerated training via CUDA, focusing on performance benchmarking and model validation.
- Technologies: Python, PyTorch, TensorFlow, CUDA, Matplotlib, Scikit-learn, Google Colab.

Machine Learning Research Intern

HamsanTech, NSTP | June 2024 - Aug 2024

Supervisor: Dr. Nazia Perwaiz, Assistant Professor

- Developed interactive business dashboards using Power BI and Tableau, extracting actionable insights from real-life industry datasets; used Excel for data cleaning, preprocessing, and exploratory data analysis.
- Built a multi-model skin cancer classification pipeline achieving 91% accuracy and AUC of 0.963 using ensemble methods including XGBoost, AdaBoost, LightGBM, SVM, and Logistic Regression.
- Technologies: Python, Scikit-learn, Pandas, Matplotlib, TensorFlow, PyTorch, Excel, Power BI, Tableau.

SELECTED RESEARCH PROJECTS

AI-Based Exam Generator using Retrieval-Augmented Generation (RAG)

2025

- Developed an end-to-end automated exam generation system using RAG architecture with Groq API and Llama 3.1, integrating document ingestion, vector embedding, semantic retrieval, and LLM-based question generation into a production-ready pipeline.
- Demonstrated applied competency in LLMs, prompt engineering, and RAG system design, directly relevant to AI research and NLP workflows.
- Tools: Python, LangChain, Groq API, Llama 3.1, Vector Databases, RAG.

FYP: ML-Based Hardware Accelerator for Real-Time Image Segmentation on FPGA

2025-2026

- Achieved 56x model compression to 570 KB with 97% pixel accuracy and 2.72W power consumption, bridging the full pipeline from ML model training and quantization to edge hardware deployment.
- Tools: Zybo Z7-20, Python, PyTorch, Vitis HLS, Xilinx Vivado.

Autonomous Driving Scene Segmentation with Lightweight U-Net

Dec 2025

- Designed a novel lightweight U-Net compressing 8.7M to 0.53M parameters while maintaining 72% IoU, 80% F1, and 96% test accuracy on the CARLA simulator dataset. Paper in progress based on this architecture.
- Tools: Python, PyTorch, OpenCV, Scikit-learn, Google Colab.

Real-Time Fabric Defect Detection on Jetson Nano

Dec 2025

- Classical computer vision pipeline using GLCM, FFT, Gabor, and statistical methods for robust defect localization on a low-power edge device; paper in progress on methodology and deployment results.
- Tools: Python, OpenCV, NumPy, SciPy, scikit-image, Jetson Nano.

Automated Waste Classification using EfficientNet

Nov 2025

- Transfer learning system using EfficientNetB0 achieving 92.9% test accuracy and macro F1 score of 0.93 across multiple waste categories.
- Tools: Python, TensorFlow, Keras, EfficientNetB0, Kaggle.

Skin Cancer Detection using Multi-Model Ensemble

2024

- Ensemble ML pipeline achieving 91% accuracy and AUC of 0.963 for single-image RGB skin cancer classification using XGBoost, AdaBoost, LightGBM, SVM, and Logistic Regression.
- Tools: Python, Scikit-learn, Pandas, Matplotlib, PyTorch, TensorFlow.

TECHNICAL SKILLS

Machine Learning and AI: Python, TensorFlow, PyTorch, Keras, Scikit-learn, NumPy, Pandas, OpenCV, SciPy, CUDA

LLMs and RAG: LangChain, Groq API, Llama 3.1, Vector Databases, Retrieval-Augmented Generation, Prompt Engineering

Data Analysis and Visualization: Excel, Power BI, Tableau, Matplotlib, Seaborn, Pandas

Development and Tools: Git/GitHub, Linux, Jupyter Notebook, Google Colab, VS Code, Kaggle, LaTeX

RELEVANT COURSEWORK

Machine Learning Specialization (3-Course Series)

deeplearning.ai

- Supervised ML, advanced learning algorithms, unsupervised learning, and reinforcement learning.

Deep Learning Specialization (5-Course Series)

deeplearning.ai

- Neural networks, hyperparameter tuning, CNNs, RNNs, LSTMs, sequence modeling, and NLP.

Deep Learning Course Series

MIT 6.S191

- Computer vision, generative modeling, transformers, LLMs, deep reinforcement learning, and AI in biology.

NUST Coursework: Machine Learning, Deep Learning, Computer Vision, Linear Algebra, Calculus, Digital Signal Processing, Signals and Systems.

HONORS AND ACHIEVEMENTS

- **Mitacs Globalink Research Internship:** Competitive, fully funded international research placement at NRC Herzberg, Canada (Summer 2026).
- **Merit Scholarship 2024-2026:** Awarded by NUST SECS for sustained academic excellence.
- **Dean's Honor List:** Recognized 4 times for outstanding academic performance.
- **Millennium Fellowship:** Selected among top 4% globally (UN Academic Impact and Millennium Campus Network).
- **Prime Minister Youth Laptop Scheme 2025:** Awarded for academic excellence.
- **Position Holder:** 3rd place in BISE Multan Pre-Engineering among 70,000+ students.
- **Intermediate Scholarship:** 100% fee waiver based on exceptional academic performance.